



LifeLine — Comprehensive Blood Donation Platform

Document: Project Proposal

Course: CSC13002 - Introduction to Software Engineering

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1. Introduction

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

1.1. Project idea

LifeLine is a web-based blood donation platform that connects donors, blood donation centers, and hospitals in one ecosystem. The system simplifies the donation process through online registration, appointment booking, donation tracking, and emergency blood request notifications.

1.2. Project statement

Current blood donation activities often rely on fragmented communication and manual procedures. Donors may struggle to find reliable information, schedule appointments conveniently, or stay informed about urgent blood shortages. In addition, organizations face difficulties in maintaining long-term engagement with donors and providing personalized care to a large number of users, resulting in lower donor retention and fewer repeat donations.

1.3. Project value

LifeLine improves the blood donation experience by making the process more accessible, efficient, and transparent. The platform helps healthcare organizations better manage and support donors throughout their donation journey. Through personalized reminders, achievement tracking, and continuous engagement features, LifeLine encourages users to donate again in the future while ensuring they receive better care and attention before and

after each donation.

2. Target Users and Environment

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

2.1. Target users

Blood donors

Individuals who want to donate blood, track their donation history, receive reminders, and stay informed about donation opportunities and emergency requests.

Blood donation centers

Organizations responsible for managing donation campaigns, appointments, donor verification, and communication with registered donors.

Hospitals

Healthcare institutions that need to submit urgent blood requests and quickly connect with eligible donors during critical situations.

2.2. Environment

- **Devices:** Smartphones, tablets, laptops, and desktop computers with internet access.
 - **Platform:** The software is intended for use on websites accessed via personal computers and laptops.
 - **Usage environment:** Hospitals, blood donation centers, mobile donation events, educational institutions, workplaces, and personal environments.
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3. Product Features (Functional Groups)

(Author: Trần Minh Triết | Reviewer: Trần Anh Kiệt | Editor: Trần Minh Triết)

3.1. User Features

3.1.1. User Account Management

- **Registration via Citizen ID (CCCD) or Passport:** Users can create an account by scanning their CCCD QR code or uploading a Passport photo. This ensures each donor profile is tied to a verified identity document, guaranteeing the authenticity of records and eliminating duplicate accounts.
- **Login:** Users securely sign in using their CCCD or Passport number paired with their registered password.
- **Profile Management:** Users can view and update personal details, including contact information and residential address, keeping their donor profile accurate and up to date.

3.1.2. Donation Booking & Location Services

- **Interactive Map and Location Discovery:** Users can explore nearby donation points and active campaigns through an integrated map interface featuring distance and availability filters. This allows donors to quickly identify the most convenient locations without manually searching multiple sources.
- **Appointment Scheduling:** Donors can select their preferred date, time slot, and location. The system displays matching campaigns, and all booking details are instantly recorded, streamlining the process and reducing administrative workload at the venue.

3.1.3. Donor Guidance & Support (Q&A AI)

- **AI-Powered FAQ:** Users can ask an AI assistant common questions about procedures, requirements, and health standards. It provides instant, accurate answers drawn from a curated medical knowledge base, eliminating the need to search static pages or wait for staff.
- **Pre- and Post-Donation Guidance:** Users receive tailored advice on preparation and recovery based on their personal health context, improving donor confidence and ensuring a safer experience.

3.1.4. News, Notifications & Communication

- **News Feed and Campaign Announcements:** A centralized feed displays the latest news, upcoming campaigns, health tips, and event announcements, keeping donors consistently informed and engaged.

3.1.5. Donation Impact & Tracking (Journey)

- **Donation History and Details:** Users access a chronological timeline of their past donations, displaying the date, location, campaign name, and blood volume donated, providing a clear overview of their contribution history.
- **Milestone Badges and Achievements:** The system automatically awards badges and experience points when predefined milestones are reached. This recognition system fosters a sense of accomplishment and encourages retention.
- **Gamification Progress Tracking:** Visual indicators (e.g., donor levels, growing virtual trees) and personalized motivational messages help users track their progress toward their next milestone, reinforcing continued participation.

3.1.6. Community (Facebook Fanpage Redirect)

- **Direct Fanpage Link:** A prominent access button redirects users to the official Facebook fanpage, allowing them to stay connected via a familiar platform without requiring a separate in-app social network.

3.2. Blood Center Features

3.2.1. Blood Donation Campaign and Management

- **Event Creation and Configuration:** Staff can create events and configure parameters such as target blood groups, location, time, and participant capacity. This streamlines planning and enforces registration limits.
- **QR Code Scanning and Verification:** Staff scan the QR code on a donor's e-ticket to instantly retrieve and verify registration details. This contactless procedure replaces manual check-ins, making verification faster and more accurate.

3.2.2. Communication and User Engagement Management

- **Content Publishing:** Staff can author, edit, and publish news articles and campaign announcements via a built-in CMS, ensuring donors have centralized access to organizational updates.
- **Emergency Announcements:** Staff can broadcast urgent blood shortage alerts to targeted donor segments, enabling rapid mass communication during critical needs.

3.2.3. Blood Inventory and Emergency Coordination Management

- **Blood Bag Inventory Tracking:** The system maintains granular records of every blood bag (blood type, quantity, collection date, storage location), providing real-time visibility over available stock.

3.3. Hospital Features

4.3.1. Emergency Blood SOS Request Management

- **SOS Request Creation:** Hospital staff generate emergency requests specifying blood type, quantity, urgency, and deadline. The system automatically validates the inputs to reduce miscommunication during crises.
- **Emergency Alert Broadcasting:** Verified SOS alerts are simultaneously broadcast to matched donors and blood centers via push notifications, SMS, and email. These alerts bypass standard rate limits to prioritize urgent outreach.

3.4. System Features (Backend & Automations)

3.4.1. User-Facing Automations

- **Pre-Donation Screening Form Generation:** The system dynamically generates digital health screening forms for users to complete prior to finalizing their appointment.
- **E-Ticket & QR Generation:** Upon successful booking, the system automatically compiles the data and generates a personalized electronic appointment ticket encoded as a QR code.

3.4.2. Blood Center-Facing Automations

- **Digital Donor Record Generation:** The system auto-generates a comprehensive digital evaluation form for each registrant. Blood Center staff utilize this centralized interface on-site to log health screening results, track donor status, and input critical medical data.
 - **SOS Request Evaluation and Prioritization:** The system intelligently evaluates incoming hospital emergency requests against current real-time inventory, automatically prioritizing competing requests based on urgency to ensure efficient resource allocation.
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4. System User Roles

(Author: Nguyễn Quốc Dương | Reviewer: Trần Anh Kiệt | Editor: Nguyễn Quốc Dương)

System Functions	Donor	Blood Center Staff	Hospital Staff	Administrator
Register Account				
Log In & Authentication				
Manage Profile				
View Donation History				
Book Donation Appointment				
Receive Notifications & SOS Alerts				
Use AI Chatbot Support				
Access Community Page				
Create Donation Campaigns				
Manage Donor Registrations				
Conduct Donor Screening				
Generate & Scan QR Check-in				
Manage Blood Inventory				
Publish News & Announcements				
Create Emergency SOS				

System Functions	Donor	Blood Center Staff	Hospital Staff	Administrator
<i>Requests</i>				
<i>Monitor SOS Request Status</i>			●	
<i>Update Blood Usage Records</i>			●	
<i>Generate Donor Impact Reports</i>			●	
<i>Manage User Accounts</i>				●
<i>Manage Roles & Permissions</i>				●
<i>Monitor System Activities</i>				●
<i>Manage System Configuration</i>				●

5. AI-Powered Feature

(Author: Trần Đức Quý | Reviewer: Trần Anh Kiệt | Editor: Trần Đức Quý)

AI-Powered Donor Assistant

An intelligent chatbot that provides personalized blood donation guidance, answers donor questions, and performs preliminary eligibility assessments based on user information and donation history.

- **Input:** User questions, health information, profile data, and donation history.
- **Output:** Personalized guidance, eligibility recommendations, and relevant action suggestions.

- **Value:** Improves donor support, reduces uncertainty, encourages repeat donations, and enables users to receive timely assistance anytime.